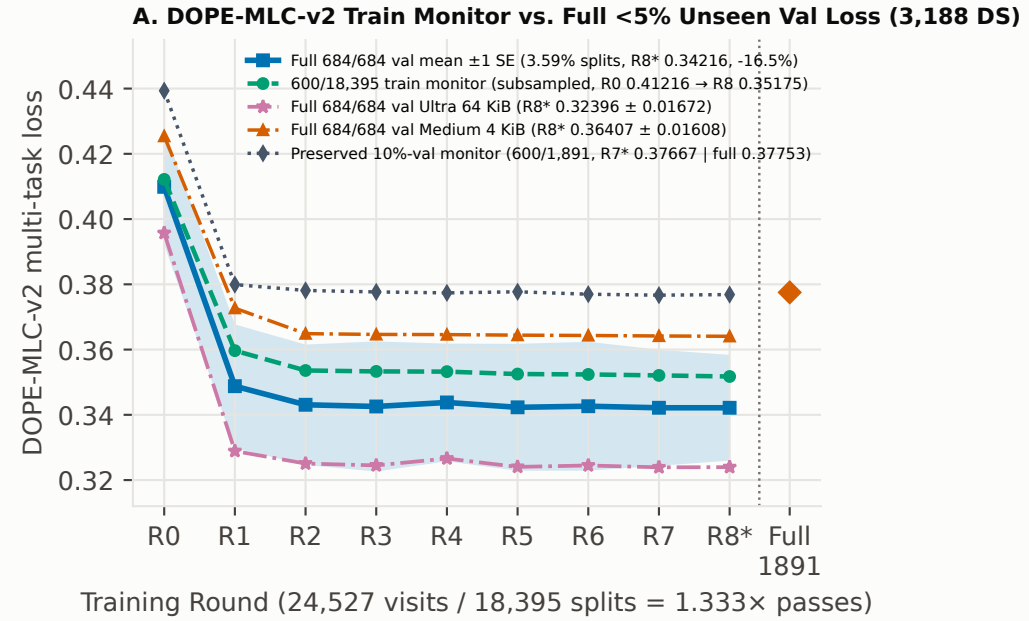
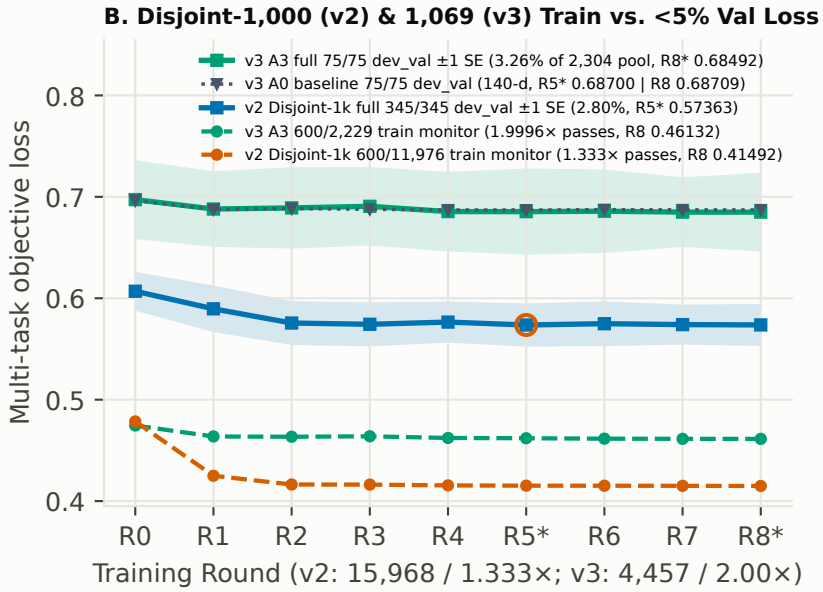
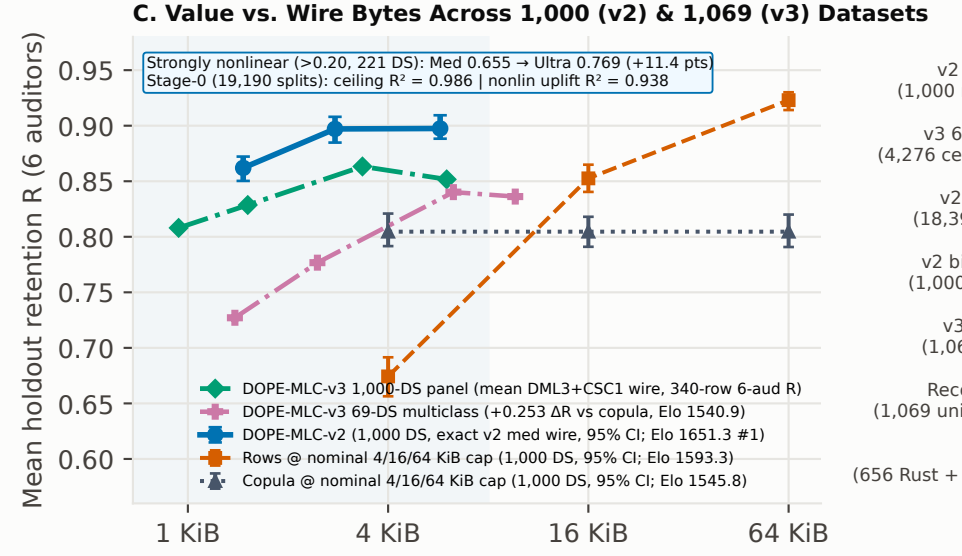
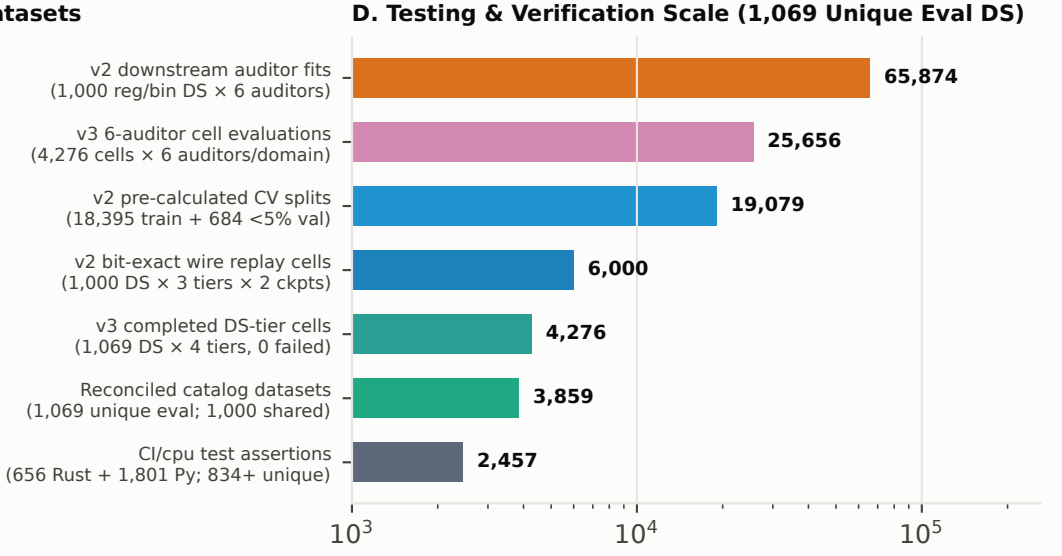

A. DOPE-MLC-v2 Train Monitor vs. Full <5% Unseen Val Loss (3,188 DS)
Full 684/684 val mean ±1 SE (3.59% splits, R8* 0.34216, -16.5%)
600/18,395 train monitor (subsampled, R0 0.41216 → R8 0.35175)
Full 684/684 val Ultra 64 KiB (R8* 0.32396 ± 0.01672)
Full 684/684 val Medium 4 KiB (R8* 0.36407 ± 0.01608)
Preserved 10%-val monitor (600/1,891, R7* 0.37667 | full 0.37753)
DOPE-MLC-v2 multi-task loss
0.44
0.42
0.40
0.38
0.36
0.34
0.32
R0
R1
R2
R3
R4
R5
R6
R7
R8*
Full 1891
Training Round (24,527 visits / 18,395 splits = 1.333× passes)


B. Disjoint-1,000 (v2) & 1,069 (v3) Train vs. <5% Val Loss
v3 A3 full 75/75 dev_val ±1 SE (3.26% of 2,304 pool, R8* 0.68492)
v3 A0 baseline 75/75 dev_val (140-d, R5* 0.68700 | R8 0.68709)
v2 Disjoint-1k full 345/345 dev_val ±1 SE (2.80%, R5* 0.57363)
v3 A3 600/2,229 train monitor (1.9996× passes, R8 0.46132)
v2 Disjoint-1k 600/11,976 train monitor (1.333× passes, R8 0.41492)
Multi-task objective loss
0.8
0.7
0.6
0.5
0.4
R0
R1
R2
R3
R4
R5*
R6
R7
R8*
Training Round (v2: 15,968 / 1.333×; v3: 4,457 / 2.00×)


C. Value vs. Wire Bytes Across 1,000 (v2) & 1,069 (v3) Datasets
Strongly nonlinear (>0.20, 221 DS): Med 0.655 → Ultra 0.769 (+11.4 pts)
Stage-0 (19,190 splits): ceiling R² = 0.986 | nonlin uplift R² = 0.938
Mean holdout retention R (6 auditors)
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
DOPE-MLC-v3 1,000-DS panel (mean DML3+CSC1 wire, 340-row 6-aud R)
DOPE-MLC-v3 69-DS multiclass (+0.253 ΔR vs copula, Elo 1540.9)
DOPE-MLC-v2 (1,000 DS, exact v2 med wire, 95% CI; Elo 1651.3 #1)
Rows @ nominal 4/16/64 KiB cap (1,000 DS, 95% CI; Elo 1593.3)
Copula @ nominal 4/16/64 KiB cap (1,000 DS, 95% CI; Elo 1545.8)
1 KiB
4 KiB
16 KiB
64 KiB
Wire bytes (v2: exact median wire; v3: mean DML3+CSC1; baselines: cap)


D. Testing & Verification Scale (1,069 Unique Eval DS)
v2 downstream auditor fits
(1,000 reg/bin DS × 6 auditors)
65,874
v3 6-auditor cell evaluations
(4,276 cells × 6 auditors/domain)
25,656
v2 pre-calculated CV splits
(18,395 train + 684 <5% val)
19,079
v2 bit-exact wire replay cells
(1,000 DS × 3 tiers × 2 ckpts)
6,000
v3 completed DS-tier cells
(1,069 DS × 4 tiers, 0 failed)
4,276
Reconciled catalog datasets
(1,069 unique eval; 1,000 shared)
3,859
CI/cpu test assertions
(656 Rust + 1,801 Py; 834+ unique)
2,457
10³
10⁴
10⁵
Verified count (log scale; distinct non-summed receipt units)